

NIBIR CHANDRA MANDAL

Graduate Research Assistant and Graduate Teaching Assistant

Ph.D. Candidate, Department of Computer Science ◊ University of Virginia, Charlottesville, Virginia.

Academic Link: [google scholar](#) , [linkedin](#) , [research gate](#) 

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SUMMARY

- Fourth-year Ph.D. student in Computer Science at the University of Virginia
- Working as a Graduate Research Assistant at Biocomplexity Institution, UVA
- Former Lecturer at the Department of CSE, AUST
- Former University-level competitive programming
- Research experience in Natural Language Processing, Computer Vision, Generative AI, and ML applications

EDUCATION

University of Virginia

August 2022 - Present

Ph.D. in Computer Science (ongoing)

Advisor: Madhav Marathe

Bangladesh University of Engineering & Technology

February 2015 - May 2019

B.Sc. in Computer Science

Advisor: Mohammed Eunus Ali

RESEARCH INTERESTS

Deep Learning – Generative modeling, foundation models, and data-efficient training methods.

Computer Vision & Multimodal Learning – Vision–language models, multimodal representation learning, and image segmentation.

Knowledge-Informed AI – Integration of symbolic knowledge and constraints into machine learning for robust and interpretable systems.

Applied AI – Development of generalizable AI frameworks transferable across domains such as remote sensing, healthcare, and security.

EXPERIENCE

Graduate Research Assistant (*August 2022 - Present*)

Department of Computer Science, University of Virginia

Research Assistant (*December 2020 - May 2022*)

[DISA Lab](#), University of Calgary

Lecturer (*July 2019 - August 2022*)

Department of Computer Science and Engineering, AUST

Frontend Software Developer (*November 2020 - May 2021*)

Dast Solution, USA

Research Assistant (*February 2018 - April 2019*)

[Data Lab](#), Bangladesh University of Engineering & Technology

Google scholar citation: 43  Research gate citation: 45 

- **Nibir Chandra Mandal**, Gursharn Kaur, Aniruddha Adiga, Madhav Marathe, Benjamin Hurt, Srinivasan Venkatramanan, Bryan Lewis. “**Optim-WISE: An Ensemble Approach for Probabilistic Forecasts Optimized via Weighted Interval Score.**” *Innovative Applications of AI Conference (IAAI) 2026 (Under Review)*
- **Oishee Bintey Hoque***, **Nibir Chandra Mandal***, Samarth Swarup, Mandy Wilson, Madhav Marathe. “**From Weak Coordinates to Structured Understanding: A Self-Supervised Framework for Discovering Functional Structures in Remote Sensing**” *Neural Information Processing Systems, 2025.(Under Review)*
- **Nibir Chandra Mandal***, **Oishee Bintey Hoque***, Samarth Swarup, Mandy Wilson, Madhav Marathe. “**IrrigationNet: A Multimodal Benchmark Dataset for Large-Scale Irrigation Type Mapping across the U.S.**” *Neural Information Processing Systems, 2025.(Under Review)*
- **Nibir Chandra Mandal***, **Oishee Bintey Hoque***, Abhijin Adiga, Samarth Swarup, Mandy L Wilson, Lu Feng, Yangfeng Ji, Miaomiao Zhang, Geoffrey Fox, Madhav Marathe. “**IrrMap: A Large-Scale Comprehensive Dataset for Irrigation Method Mapping.**” Funded by: Agricultural AI for Transforming Workforce and Decision Support (AgAID) award No. 2021-67021-35344. *31st SIGKDD Conference on Knowledge Discovery and Data Mining, 2025.*
- **Nibir Chandra Mandal***, **Oishee Bintey Hoque***, Samarth Swarup, Mandy Wilson, Madhav Marathe. “**Knowledge-Guided Deep Learning for Advanced Irrigation Method Mapping**” *International Joint Conference on Artificial Intelligence, 2025.*
- **Nibir Chandra Mandal**, Gursharn Kaur, Aniruddha Adiga, Madhav Marathe, Srinivasan Venkatramanan, Bryan Lewis. “**An Exploration of Trained Ensemble Models for Epidemic Forecasting Based on Weighted Interval Scores.**” *International Symposium on Forecasting, 2023*
- **Nibir Chandra Mandal**, GM Shahariar, Md Tanvir Rouf Shawon. “**Effectiveness of Transformer Models on IoT Security Detection in StackOverflow Discussions.**” *Proceedings of International Conference on Information and Communication Technology for Development. Studies in Autonomic, Data-driven and Industrial Computing. Springer, Singapore.* 
- Nipun Paul, Ishmam Faruki, Mutakabbirul Islam Pranto, Md Tanvir Rouf Shawon, **Nibir Chandra Mandal** “**Bengali-English Neural Machine Translation Using Deep Learning Techniques.**” *2023 International Conference on Electrical, Computer and Communication Engineering (ECCE), Chittagong, Bangladesh, 2023, pp. 1-6.* 
- G. M. Shahariar Shibli, Md. Tanvir Rouf Shawon, Anik Hassan Nibir, Md. Zabed Miandad & **Nibir Chandra Mandal**. “**Automatic back transliteration of Romanized Bengali (Banglish) to Bengali.**” *Iran J Computer Science 6, 69–80 (2023).* 
- **Nibir Mandal**, Gias Uddin “**An empirical study of IoT security aspects at sentence-level in developer textual discussions.**” *Information and Software Technology 150 (2022): 106970.* 
- Nawrin Tabassum, Monjure Mowla, Kazi Fuad Bin Akhter, Md Tanvir Rouf Shawon, & **Nibir Chandra Mandal**. “**An Empirical Study of Pre-Trained CNN Models on COVID-19 CT Scan Images.**” *2022 4th International Conference on Sustainable Technologies for Industry 4.0 (STI). IEEE, 2022.* 
- Arman Sharker, Mohammad Abdur Rafi Farhab, Tahrima Akter Tamanna, Umma Rumman, Md Tanvir Rouf Shawon, & **Nibir Chandra Mandal**. “**A Cross-Corpus Deep Learning Approach to Social Media Emotion Classification.**” *2022 4th International Conference on Sustainable Technologies for Industry 4.0 (STI), Dhaka, Bangladesh, 2022, pp. 1-6.* 
- Fatima Juairiah, Mostafa Mahatabe, Hasan Bin Jamal, Aysha Shiddika, Tanvir Rouf Shawon & **Nibir Chandra Mandal**. “**Stock Price Prediction: A Time Series Analysis.**” *2022 25th In-*

ternational Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2022, pp. 153-158. 

- Aysha Gazi Mouri, Purnendu Talukder, Tanvir Rahman Anik, Ifti Sam Ibn Rahman, Sajib Kumar Saha Joy, Md Tanvir Rouf Shawon, Farzad Ahmed & Nibir Chandra Mandal. “An Empirical Study on Bengali News Headline Categorization Leveraging Different Machine Learning Techniques.” *2022 25th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2022, pp. 312-317.* 
- Sajib Kumar Saha Joy, Farzad Ahmed, Al Hasib Mahamud & Nibir Chandra Mandal. “An Empirical Studies on How the Developers Discussed about Pandas Topics.” *Machine Intelligence and Emerging Technologies. MIET 2022. Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, vol 491.* 
- Nibir Chandra Mandal, Tashreef Muhammad, GM Shahariar “Can Transformer Models Effectively Detect Software Aspects in StackOverflow Discussion?.” *Machine Intelligence and Emerging Technologies. MIET 2022. Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, vol 491.* 

CONFERENCE PRESENTATIONS & POSTERS

- Nibir Chandra Mandal, Oishee Bintey Hoque, Abhijin Adiga, Madhav Marathe. “KIIM: Deep Learning Based Knowledge Informed Irrigation Mapping from Remote Sensing.” *UVA Research Computing Exhibition, 2025*
- Nibir Chandra Mandal, Gursharn Kaur, Aniruddha Adiga, Madhav Marathe, Srinivasan Venkatramanan, Bryan Lewis. “Optim-WISE: An Ensemble Approach for Probabilistic Forecasts Optimized via Weighted Interval Score.” *Poster in Models of Infectious Disease Agent Study (MIDAS) Annual Meeting, 2024* 
- Nibir Chandra Mandal, Gursharn Kaur, Aniruddha Adiga, Madhav Marathe, Srinivasan Venkatramanan, Bryan Lewis. “Optimizing Ensemble Forecasts for Epidemic Disease Prediction through Computational Enhancement.” *Finalist in UVA Research Computing Exhibition, 2024*
- Nibir Chandra Mandal, Gursharn Kaur, Aniruddha Adiga, Madhav Marathe, Srinivasan Venkatramanan, Bryan Lewis. “An Exploration of Trained Ensemble Models for Epidemic Forecasting Based on Weighted Interval Scores.” *Lightening Talk in International Symposium on Forecasting, 2023*
- Nibir Chandra Mandal, Tomal Mojumdar, Eunus Ali “Efficient Route Caching and Route Finding” *Poster in 5th International Conference on Networking, Systems, and Security at BUET, 2018*  

PROJECTS

- **Irrigation Mapping** address the challenge of mapping irrigation methods across diverse agricultural landscapes, a critical task for water resource management hindered by limited data and complex regional variations.
- **Epidemic Forecasting** is funded by the Virginia Department of Health (VDH). The objective of this project is to forecast epidemic diseases to facilitate policymakers for better policy enforcement.
- **Synthetic Data Generation** project aims to generate artificial datasets that mimic real-world data while preserving its statistical properties and patterns for the health sector.
- **Intelligent Bengali Text Translation** Focused on leveraging neural machine translation and deep learning techniques to enhance Bengali-English translation quality and address linguistic challenges effectively.

- **Empirical Analysis of Software Discussions** Conducted comprehensive studies on developer discussions from platforms like StackOverflow, with an emphasis on IoT security, software trends, and developer communication, utilizing advanced transformer-based models.

PROFESSIONAL SERVICE

- Peer Reviewer, KDD 2025 (Knowledge Discovery and Data Mining) (One of top conference in Computer Science)
- Peer Reviewer, KDD 2024 (Knowledge Discovery and Data Mining) (One of top conference in Computer Science)
- Organizing Committee, ICCIT 2020 (The 23rd International Conference on Computer and Information Technology)
- Seminar Organizing Committee, AUST (2019 - 2021)
- Invigilator Committee, AUST (Fall-2020)
- Outcome-Based Education (OBE) Subcommittee, Department of Computer Science, AUST (2021)

FELLOWSHIP, HONOR & AWARDS

- Ph.D. fellowship from the Computer Science Department, University of Virginia
- Ranked 1st in section merit list (among 60 students), BUET
- Dean's List of Honor (based on B.Sc. result)
- University Merit List, BUET 2015

LEADERSHIP POSITIONS

- Mentored two undergraduate research assistants on machine learning projects (2025)
- Elected and am serving as a Leadership Chair at UVA CSGSG-2025
- Undergraduate admission sub-committee member of Spring 2020 semester at AUST
- Undergraduate academic advisor of the Fall 2019–Spring 2021 Semester
- IT Support Subcommittee member at the 23rd International Conference on Computer and Information Technology (ICCIT, 2020)
- Festival organizing committee member, BUET (CSE Fest - 2019)
- Outcome-Based Education Committee member, CSE, AUST (2021)

REFERENCES

Madhav Marathe

Distinguished Professor in Biocomplexity, Biocomplexity Institute
University of Virginia
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Rifat Shahriyar

Professor
Department of Computer Science and Engineering, BUET
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